Eldis YMERAJ

AI & Machine Learning Engineer

eldisymeraj0@gmail.com | +33 7 83 88 39 26 | LinkedIn | GitHub

EDUCATION

Master's in Artificial Intelligence *Sept. 2024 – Sept. 2026*
University of Caen Normandy Caen, France

- Expected graduation in September 2026.

Bachelor's in Computer Science *Sept. 2021 – May 2024*
University of Clermont Auvergne Clermont-Ferrand, France

SKILLS

- **Programming Languages:** Python, Java, C, C++
- **Machine Learning & Computer Vision:** Object Detection, Pose Estimation, YOLO, Explainable AI (XAI), PCA, NMF
- **Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, OpenCV, NumPy, Pandas
- **NLP & Generative AI:** RAG, Transformers, Hugging Face, CamemBERT, Embeddings
- **Data Engineering & Cloud:** AWS (EC2, S3), SQL, PostgreSQL, MongoDB, Apache Hop, Data Visualization (Matplotlib, Plotly)
- **Development Tools:** Git, GitHub, Linux, Conda, Jupyter Notebook, Google Colab, VS Code
- **Web Technologies:** HTML, CSS, JavaScript

PROFESSIONAL EXPERIENCE

Airbus SAS *March 2026 – Sept. 2026*
Artificial Intelligence Intern – Trustworthy AI Toulouse, France

- Adapted an Explainable AI method to a multi-scale YOLO-NAS Pose detection model using PyTorch to identify the visual features influencing model detections.
- Developed an experimental protocol to extract, select, and analyze visual representations, assessing their quality, stability, and ability to preserve information relevant to detections.
- Quantitatively evaluated detection fidelity after integrating the explainability method using detection and spatial matching metrics, achieving **90.5% precision** and a **PR-AUC of 0.776**.
- Analyzed model behavior across different runway distances to identify the most influential visual information and study how it evolves throughout the approach.

Responsibio *May 2025 – Aug. 2025*
Data Science Intern Caen, France

- Collected and prepared mouse vocalization recordings and associated metadata to build a dataset for animal welfare classification.
- Developed an audio classification pipeline using mel-spectrograms, feature extraction, PCA dimensionality reduction, and a linear SVM classifier.
- Evaluated the model using 5-fold cross-validation, reducing 2048 features to 58 while retaining 86 % of the variance and achieving approximately **79 % correct predictions**.

PROJECTS

Handball Video Detection and Tracking *Sept. 2024 – Dec. 2024*
Python, PyTorch, YOLOv8, OpenCV, ByteTrack, Homography

- Created and annotated datasets for players, goalkeepers, referees, the ball, and court keypoints, then trained specialized YOLOv8 detection models.
- Developed a video tracking pipeline combining ByteTrack, ball filtering and interpolation, with homography-based projection of player and ball positions onto a 2D court.

Automatic Annotation of Clinical Narratives *Sept. 2025 – Mar. 2026*
Python, PyTorch, Hugging Face Transformers, CamemBERT, Pandas

- Developed an NLP system to automatically identify episodic and semantic memory spans at token level in French clinical narratives.
- Fine-tuned and compared CamemBERT, ModernCamemBERT, and DeBERTa using patient-level cross-validation to evaluate generalization while preventing data leakage.

ADDITIONAL INFORMATION

Languages: Albanian (Native), French (C2), English (C2) **Interests:** Football, travel, cultural discovery